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Market module

Market keeps the stocks the app offers and their prices. A background thread asks a market data provider for each stock's price once a minute while the market is open, and for 15 minutes after it closes.

Folder	`src/backend/app/market/`
Tables	`assets`, `prices`, `provider_requests`
Routes	`GET /v1/assets`, `GET /v1/market/usage`
Uses	Only the kernel
Used by	trading
Issue	#187

Files

Other modules read market only through `queries.py`, which never writes, and `types.py`. The providers, the poller and the calendar are market's own. `plugin.py` adds the routes, and starts and stops the poller with the app.

Where prices come from

- **Strategy:** every provider has one method, `get_quote(symbol)`, which returns a `Quote`. The poller only calls `get_quote`, so it works the same with any provider.
- **Adapter:** `FinnhubProvider` turns Finnhub's quote API into a `Quote`, with exact `Decimal` prices. It turns Finnhub's problems into a `MarketDataError`, such as `RateLimitedError` for a 429 or `UnknownSymbolError` for a reply of zeros, so the poller records what happened without knowing the provider. These never reach the phone.
- **Fake:** `FakeProvider` makes up prices, for development and tests. Each symbol starts at the same price every run, from 20 to 500, moves by up to 0.5% on each call, and is quoted at the current time.
- **Factory:** `build_provider(settings)` returns the provider that `MARKET_PROVIDER` names, `fake` by default. An unknown name is refused, never replaced by made-up prices, which would be shown as real.

The Finnhub key stays on the server (SEC-33). It is `FINNHUB_API_KEY` in the server's `.env`, which git ignores. `ProviderSettings` keeps it as a `SecretStr`, so printing the settings hides it. `FinnhubProvider` sends it in a header, never in the URL, so no logged URL or error shows it. No route sends it, and `provider_requests` never stores it. Finnhub gets only the stock's symbol (SEC-11).

To add a provider, without changing any code outside `providers/`: 1. Write a class in `app/market/providers/` that names `MarketDataProvider` as its base and has `get_quote(symbol)`. One without `get_quote` fails as soon as it is created. 2. Add its name to `MARKET_PROVIDER` in `providers/settings.py` and to `.env.example`. 3. Add it to `build_provider`.

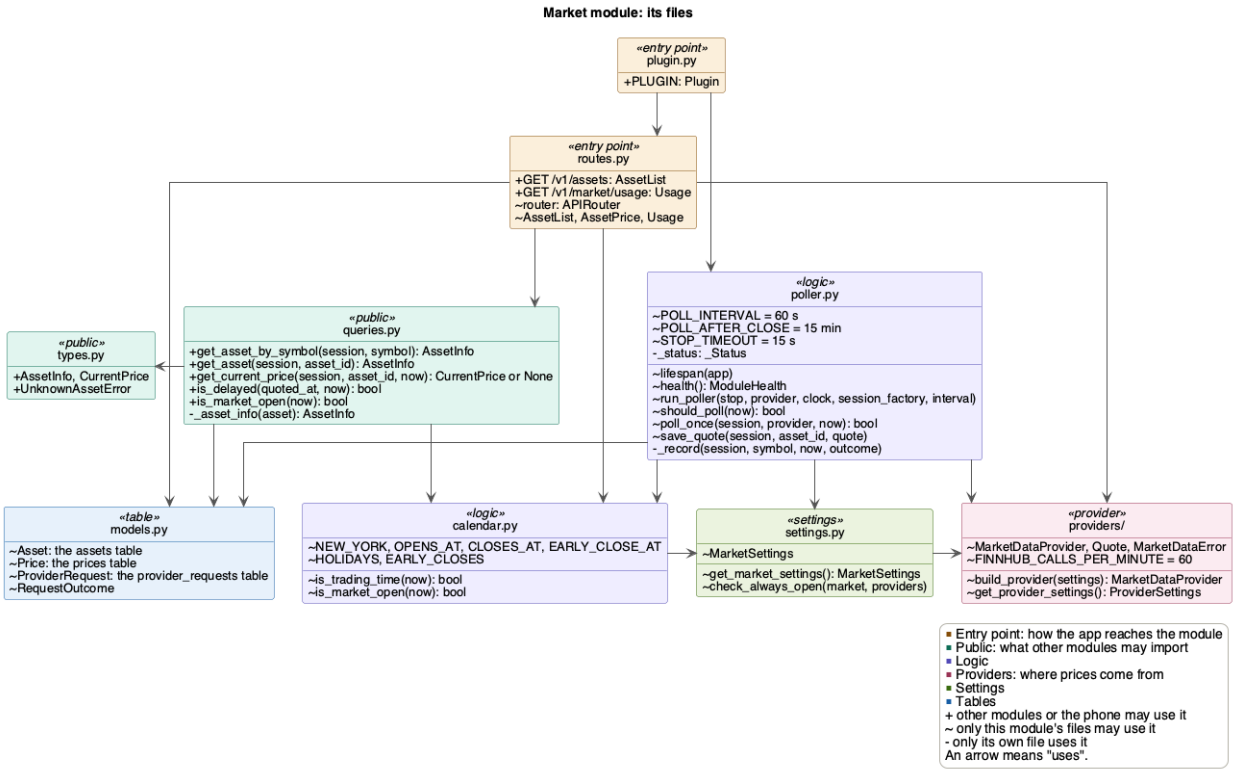


Figure 1: UML class diagram of the market module's files

The poller

`plugin.py` hands the kernel `poller.lifespan`, which runs the poller's thread while the app runs. - **Starting and stopping:** `lifespan` reads the settings first, so bad ones, such as `MARKET_PROVIDER=finnhub` without a key, stop the app from starting. It starts the thread only while `MARKET_POLLER_ENABLED` is on, which is off unless `.env` turns it on (`.env.example` does), and off in tests. When the app stops, it waits up to 15 seconds for the thread to finish. - **When:** every 60 seconds while the market is open, and for 15 minutes after it closes, since Finnhub's closing price can arrive a few minutes late. - **A poll:** `poll_once` asks for each tradable stock's price and saves it. One stock's error doesn't stop the others, but the provider's request limit stops the poll. Every call is saved in `provider_requests` with how it ended. - **Saving a price:** once per stock and quote time. An exact repeat is skipped, and a corrected price replaces the old one. Each save has its own savepoint, so a price the database refuses undoes only itself. - **Committing:** each poll has its own session and commits once, the one exception to rule 4. An error is logged and the next poll tries again, so one bad minute never stops the thread.

A poll fails if it raises an error or misses any stock's price, so a provider that is down, or a wrong key, shows up even though no request sees an error. `health` reports it in `GET /v1/health`, under `modules.market`:

`detail`	`status`	When
`poller off`	`ok`	`MARKET_POLLER_ENABLED` is off, or the app has stopped
`poller stopped`	`degraded`	The thread ended while the app still runs
`last poll failed`	`degraded`	The latest poll failed. It shows until the next poll, which may be the next trading day

detail	status	When
none	ok	The thread runs and its latest poll, if any, saved every price

Market hours

calendar.py knows when the New York Stock Exchange is open: 9:30 to 16:00 New York time on weekdays, to 13:00 on the days in EARLY_CLOSES, and never on the days in HOLIDAYS. Both list 2026 and 2027, so add the next year's dates each December. A time is turned into New York time first, so daylight saving time is handled, and a time without a zone is refused.

is_trading_time(now) gives the real hours. is_market_open(now) is what the app uses: the real hours, or always open while MARKET_ALWAYS_OPEN is on, to develop in the evening. That only works with MARKET_PROVIDER=fake. With a real provider the app refuses to start, so real prices are never shown as live while the market is closed.

When a price is delayed

A price is delayed when the market is open and the provider set it more than QUOTE_STALENESS_LIMIT (15 minutes, in app/core/constants.py) ago (FR_MktPriceAge). It goes by quoted_at, when the provider set the price, not obtained_at, when the server got it, so an old quote is delayed even if the server only just got it. While the market is closed, no price is delayed, since the last one stays the same until it opens.

Trading refuses a buy at a delayed price, but still values a holding at it.

Routes

Neither route needs a user.

GET /v1/assets sends every tradable stock, sorted by symbol, with its newest price and delayed, and whether the market is open (FR_MktRealPrices, FR_MktPriceAge). A price is text with 4 decimals, such as "84.0000", in USD. Before a stock's first price, its price and both times are null.

GET /v1/market/usage counts the server's calls to the provider, to check they stay within its request limit (SC_MarketData). It gives the calls in the last minute, in the busiest minute today and in all of today, a New York calendar day. limit_per_minute is 60, Finnhub's free plan. A poll makes one call per tradable stock, so the 15 stocks take 15 a minute.

Records, errors, settings and tables

A stock's rows in prices are its price history, and its newest price is the one with the latest quoted_at (FR_MktRealPrices). Every price is in USD, and assets and prices refuse any other currency (FR_MktOneCurrency, SEC-27). The database sets obtained_at when it saves a price.

Design patterns

- **Plug-in:** plugin.py adds market's routes, the poller's lifespan and its health. The kernel doesn't change.
- **Strategy, adapter and factory:** MarketDataProvider, FinnhubProvider and build_provider, as above.
- **Dependency injection:** the routes get the session and the clock through Depends. run_poller is given its provider, clock and session factory, FakeProvider its clock and FinnhubProvider its HTTP client, so tests hand in stand-ins.
- **Read-only records:** AssetInfo and CurrentPrice are frozen copies, never rows.

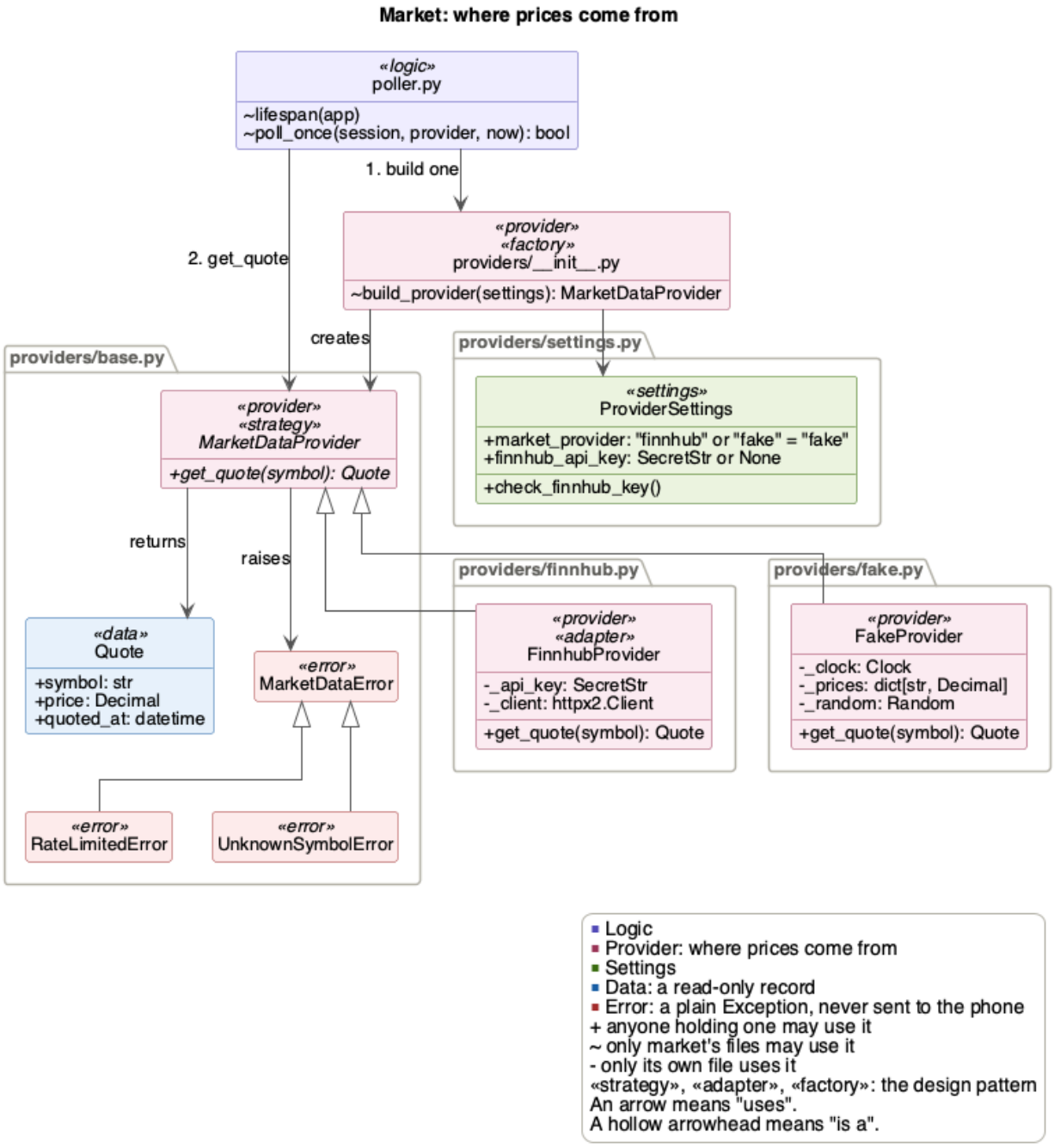


Figure 2: UML class diagram of the market data providers

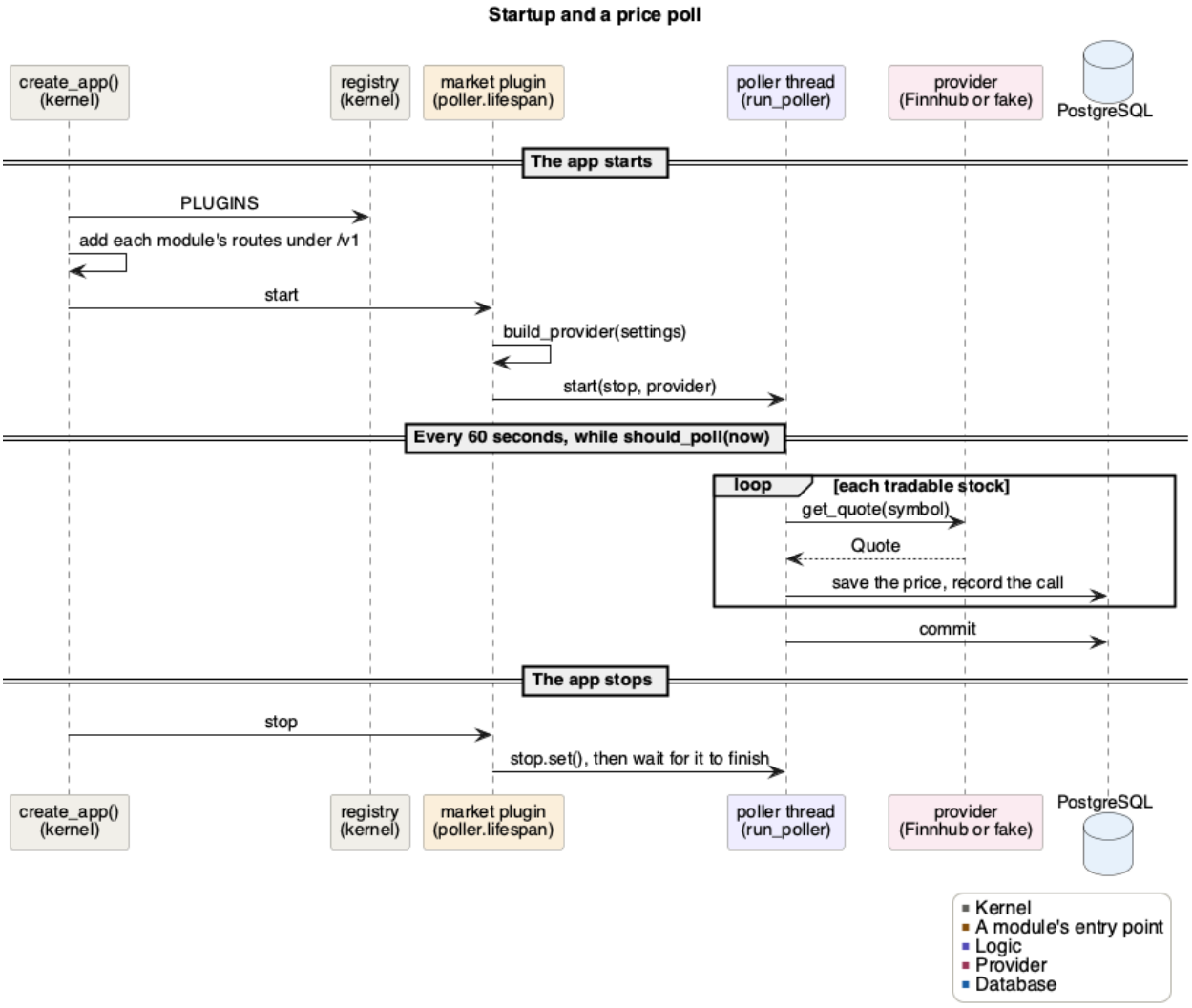


Figure 3: UML sequence diagram of startup and a price poll

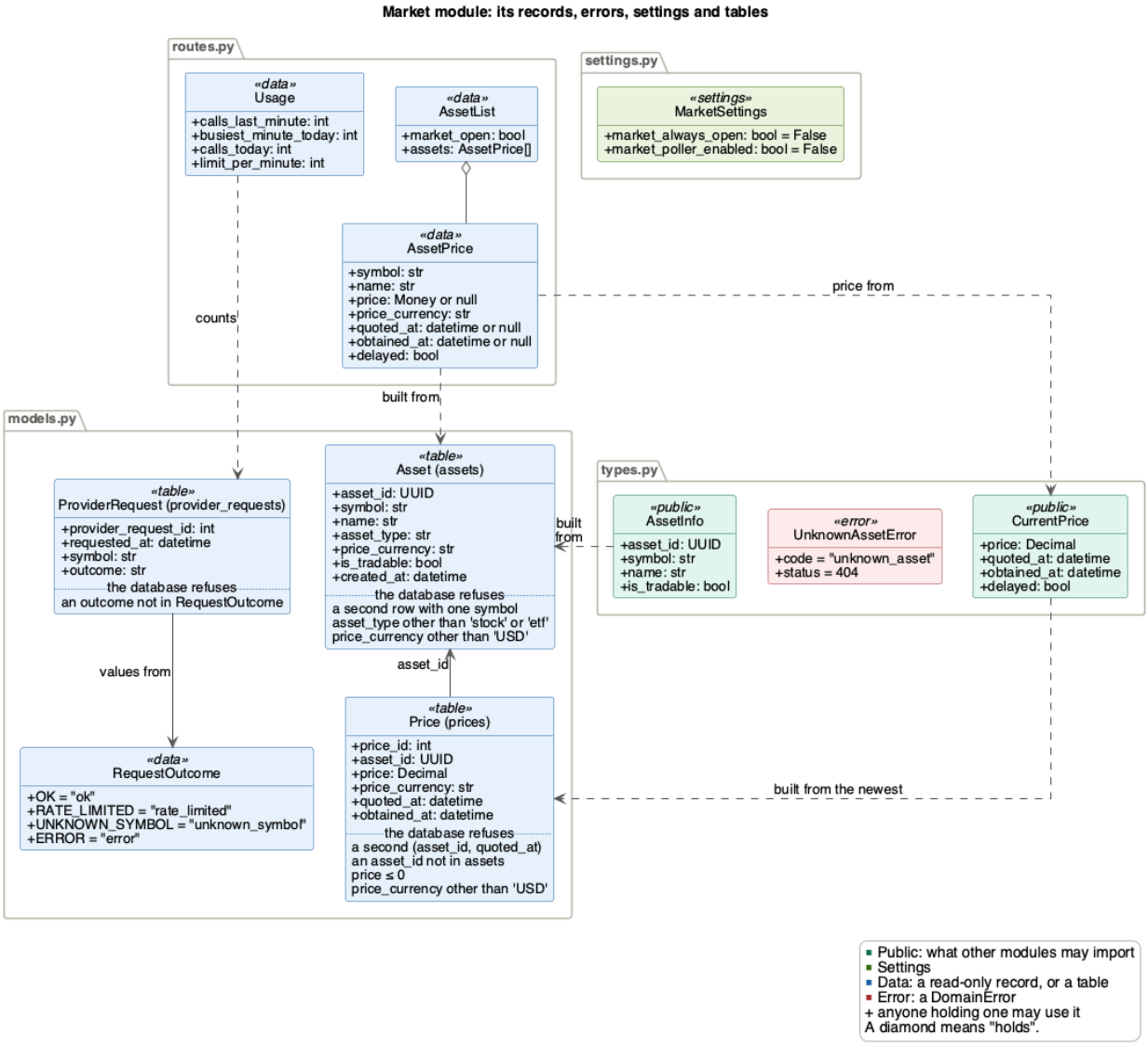


Figure 4: UML class diagram of market's records, errors, settings and tables

Not built in the PoC

- Most of the stocks. Migration `0002` adds 15, and FR_MktAssetCount asks for `MIN_TRADABLE_STOCKS` (50).
- Adding or removing a stock without a new release (FR_MktUpdateAssets), and selling every user's shares of a removed one (FR_MktRemovedHoldings). `assets.is_tradable` is there, but nothing changes it yet.
- Market scenarios, which replay past prices (FR_MktScenario).

Tests

File	What it checks
`test_market_calendar.py`	The exchange's hours, holidays and early closes, time zones, and `MARKET_ALWAYS_OPEN` only with the fake provider
`test_market_providers.py`	`build_provider`, the Finnhub key setting, a provider without `get_quote`, and the fake provider's prices
`test_finnhub.py`	The Finnhub adapter against a fake Finnhub: exact prices, the key kept out of the URL, and each error
`test_market_poller.py`	`poll_once`: a price for each tradable stock, failures, the request limit, repeats and corrections, the call log
`test_market_poller_thread.py`	When it polls, the app starting and stopping the thread, bad settings, and `health`
`test_market_queries.py`	The queries: read-only copies, unknown stocks, the newest price, and when a price is delayed
`test_market_routes.py`	`GET /v1/assets` and `GET /v1/market/usage`
`test_assets_table.py`	The 15 starting stocks, and the rows the `assets` table refuses
`test_prices_table.py`	Prices saved exactly with both times, the rows the `prices` table refuses, each quote once